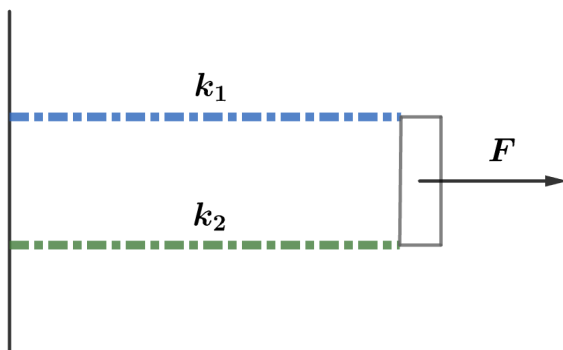


2018B F=ma Exam: Problem 18

Kevin S. Huang



Recall the effective spring constant k of two springs in parallel is given by:

$$k = k_1 + k_2$$

At the effective spring length l , the two spring forces balance (one spring is stretched while the other is compressed):

$$k_1(l - l_1) = k_2(l_2 - l)$$

$$l = \frac{k_1 l_1 + k_2 l_2}{k_1 + k_2}$$

Thus, the answer is B.